

Integrated Robotic Actuator Technical Design Document

Electromechanical architecture, control design, verification, and integration plan

PROJECT

Integrated Robotic Actuator

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Abstract

This document details the design development of the Node actuator mechanical housing, integrated with the cycloidal reducer, based on requirements from NT-N-OVR-000, and research findings from NT-N-MH-000 and NT-N-CR-000. It includes derivation of design constraint regions based on requirements, and the final design solution with justification.

Keywords: robotic actuator, BLDC, embedded control, torque sensing, mechatronics

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REVISION HISTORY

Rev.	Date	Author	Change Description
A	2026-06-25	NervTech Engineering	Initial technical document class demonstration.

1 SYSTEM OVERVIEW

System Summary

Architecture	Adhesive bonded stator and rotor assembly with integrated cycloidal reducer and space for embedded control electronics.
Control	Integrated PCB for FOC control, communication and power supply via USB-C or Molex.
Sensing	Hall effect sensors for rotor position and speed feedback, with additional quadrature encoder for higher resolution.
Safety	Low reduction ratio for high backdrivability, and vibration compensation in cycloidal drive for less transfer to fixture.

2 REQUIREMENTS

There are several goals for Node that drive the requirements and design of the actuator.

2.1 NT-N-G1

From NT-N-OVR-000, NT-N-G1 has the following goal and success criteria:

- Goal: "The actuator should provide high precision torque and position control in high motion frequency applications."
- Success Criteria: "The actuator should be able to move standard payloads up to 20kgs at frequencies comparable to human motion."

The components chosen must be able to account for failure modes induced by the stresses produced by the output load of up to 200N, applied at lever arms up to 0.25m, at frequency bandwidths up to 20Hz. This interpretation of "frequencies comparable to human motion" is derived from [1], which shows that human motion is typically in the range of 0.5Hz to 20Hz, however extended use of the upper limit is rarely reached. The lever arm of 0.25m is derived from the half length of a human arm, which is approximately 0.6m, with the 200N payload derived from the weight of a 20kg object, which would be approximately 196N.

The cycloidal speed reducer has inherent vibration characteristics, which must be accounted for in the design of the housing to ensure that the vibrations are not transferred to the fixture. Since the cycloidal speed reducer also has inherent hertzian contact stresses, the housing must be able to withstand these stresses without failure.

For the high accuracy and precision postional requirement, the overall backlash in the system should be kept to a maximum of 4 arcminutes, which would give a maximum positional error of 0.2525mm at a 0.25m lever arm, which is acceptable for the intended applications of the actuator.

Torque ripple must also be kept to a minimum to meet the high accuracy and precision torque requirement, which is defined as a maximum of 0.5% of the rated torque, which is 0.25Nm for the actuator.

2.2 NT-N-G2

From NT-N-OVR-000, NT-N-G2 has the following goal and success criteria:

Goal: "The actuator should be reusable and easily integratable into a variety of frames for robotic systems."

Success Criteria: "The actuator should be able to be mounted and dismounted with conventional tools from any of the standard NervTech linkages and/or frames."

The housing should be able to be mounted via standardised metric machine screws, with the fasteners being able to be tightened with a standard hex key or torx key. The fasteners must also be able to withstand the maximum torque applied to the actuator, which is 50Nm, without loosening or failing.

The output flange must have the same mounting pattern as the mounting base, as this pattern will provide the basis for all future linkages and actuators in the Node family of products, and will allow for easy integration of the actuator into a variety of frames for robotic systems.

2.3 NT-N-G7

From NT-N-OVR-000, NT-N-G7 has the following goal and success criteria:

Goal: "The actuator should be durable and reliable for long term use."

Success Criteria: "The actuator should be able to operate for at least 10,000 hours of continuous operation without failure, and should be able to withstand standard industrial environmental conditions including dust and high moisture."

The housing will be exposed to large temperature variations from stator heating, and as such it must be able to manage this heat long term to prevent failure. The housing will also be highly vulnerable to failure from metallic dust and contaminants unless sealed to an appropriate IP rating, which will be determined based on the final design solution. This may prevent a hollow shaft design, as this would require a large opening in the housing to allow for the shaft to pass through, which would compromise the sealing of the housing.

Since cycloidal speed reducers have inherent hertzian contact stresses, the material selection for the cycloidal plates and rollers must be appropriate to prevent failure from these stresses, and the housing must be able to withstand the forces produced by these stresses without failure (including increase in backlash and torque ripple over time). Cycloidal reducer bearings must also be chosen to withstand the forces produced by the output load at the maximum frequency of 20Hz, and the housing must be able to withstand the forces produced by these bearings without failure.

Table 1: Requirements

ID	Type	Requirement	Driving Goal	Verification
NT-N-IRQ1	Performance	The actuator shall withstand an output load of up to 200N applied at a lever arm of up to 0.25m at motion frequencies up to 20Hz without mechanical failure, including creep, plastic deformation, and fatigue.	NT-N-G1	Structural analysis and dynamic load test
NT-N-IRQ2	Vibration	The actuator housing shall limit the transfer of vibration generated by the cycloidal speed reducer to the external fixture.	NT-N-G1	Vibration test; acceptance criterion TBD
NT-N-IRQ3	Structural	The actuator housing shall withstand the reaction forces generated by Hertzian contact within the cycloidal speed reducer without structural failure.	NT-N-G1	Stress analysis and load test
NT-N-IRQ4	Precision	The total mechanical backlash of the actuator shall not exceed 4 arcminutes.	NT-N-G1	Backlash measurement test
NT-N-IRQ5	Precision	The actuator torque ripple shall not exceed 0.5% of rated torque, corresponding to 0.25Nm at the 50Nm rated torque.	NT-N-G1	Torque ripple measurement
NT-N-IRQ6	Interface	The actuator housing shall be mountable and removable using standard metric machine screws tightened using conventional hex or Torx tools.	NT-N-G2	Inspection and assembly test
NT-N-IRQ7	Structural	The actuator mounting fasteners shall withstand the maximum actuator torque of 50Nm without loosening or mechanical failure.	NT-N-G2	Fastener analysis and proof-load test
NT-N-IRQ8	Interface	The actuator output flange shall use the same mounting pattern as the actuator mounting base.	NT-N-G2	Dimensional inspection
NT-N-IRQ9	Reliability	The actuator shall be capable of at least 10,000 hours of continuous operation without failure.	NT-N-G7	Reliability analysis and endurance testing
NT-N-IRQ10	Environmental	The actuator housing shall protect internal components from dust and high-moisture industrial environments using an appropriate enclosure sealing level.	NT-N-G7	IP environmental test; required rating TBD

Table 2: Requirements

ID	Type	Requirement	Driving Goal	Verification
NT-N-IRQ11	Thermal	The actuator housing shall manage heat generated by the stator during continuous operation such that thermal loading does not cause component or housing failure.	NT-N-G7	Thermal analysis and continuous-load thermal test
NT-N-IRQ12	Durability	The cycloidal plates and roller pins shall use materials capable of withstanding the expected Hertzian contact stresses over the required operating life without failure.	NT-N-G7	Hertzian stress analysis and material verification
NT-N-IRQ13	Durability	The cycloidal reducer bearings shall withstand the forces generated by the maximum specified output load at operating frequencies up to 20Hz without failure.	NT-N-G7	Bearing load and life analysis
NT-N-IRQ14	Structural	The actuator housing shall withstand the reaction forces transmitted by the cycloidal reducer bearings throughout the required operating life without structural failure.	NT-N-G7	Structural analysis and endurance test
NT-N-IRQ15	Durability	Wear and deformation of the cycloidal reducer and its supporting structure shall not cause backlash or torque ripple to exceed their specified limits over the required operating life.	NT-N-G7	Endurance test with pre- and post-test backlash and torque-ripple measurement

3 DESIGN

3.1 Rotor Cup Design

3.1.1 Bond Line Specifications

The rotor cup design must be mounted to the rotor such that the rotor cup does not slip or rotate relative to the rotor during operation, as this would cause a loss of torque transmission and potentially damage the actuator (NT-N-IRQ1). From research done in NT-N-MH-000, adhesive bonding using retaining compounds will be the preferred method to accomplish this goal. The following retaining compound has been identified as a suitable candidate for this application:

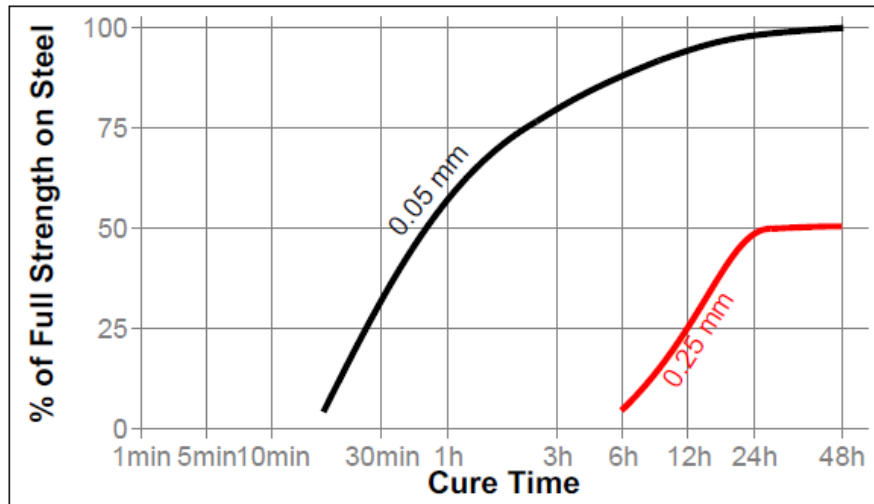


Figure 1: Bond Line Thickness vs. Cure Strength for Retaining Compounds

From the data in Figure 1, it can be seen that the bond line thickness must be kept to a minimum of 0.05mm, and a maximum of 0.25mm, to ensure the bond line strength is between 50% and 100%, assuming steel on steel bonding. The closer the bond line thickness is to 0.05mm, the higher the bond line strength will be, and the more torque can be transmitted through the bond line without failure. From the WK8115 datasheet specification (see NT-N-MH-000), the tolerance of the rotor back iron outer diameter is ± 0.1 mm, which already exceeds the minimum bond line thickness of 0.05mm.

3.1.2 Rotor Cup Tolerances

The rotor cup tolerance must be within ± 0.025 mm to ensure that the bond line thickness has a maximum thickness of 0.25mm, which in the worst case reduces the bond line strength to 50% of the maximum strength on steel. This is suitable for the application, as the maximum torque transmitted through the bond line is 5.83Nm from NT-N-MH-000, and the breakaway torque at 50% is at a minimum 10Nm after curing.

It must also be located such that its central axis is aligned with the rotor axis, as any misalignment would cause uneven air gaps (see NT-N-MH-000 for preliminary research), leading to torque ripple and potential damage to the actuator (NT-N-IRQ1) (NT-N-IRQ5). Slots for locating pins can be machined onto the backiron at 180° separation, and corresponding slots can be machined onto the rotor cup to accommodate these pins (NT-N-IRQ1) (NT-N-IRQ5).

3.1.3 Rotor Cup Cycloidal Integration

From the research done in NT-N-CR-000, to integrate an epitrochoidal-pinwheel speed reducer into the actuator (NT-N-IRQ1), 2 epitrochoidal profiles in 180° phase separation must be machined and placed on cranks, also with 180° phase separation. The 2 epitrochoidal profiles reduce the vibration from the eccentric orbit of one profile (NT-N-IRQ2). To make the stackup of rotor cup - eccentric shaft - bearing - reducer as compact as possible,

the bearings used must have minimal thickness while being able to withstand the overturning moment forces of the cycloidal output, and the radial forces from their respective eccentric orbits (NT-N-IRQ13). The cycloidal discs must furthermore be machined of materials that minimise thickness while withstanding all hertzian contact stresses, and output load stresses (NT-N-IRQ12).

3.1.4 Output Force Analysis

According to NT-N-IRQ1, the actuator should be able to apply 200Nm at a lever arm of 0.25m, up to 50Hz. Resisting axial forces as a result of moving loads is not specified in the requirements and is therefore not included in the load model. Static axial forces will, however, be considered as the actuator may be mounted in a variety of configurations, which may include resistance of a payload's weight when mounted parallel to the ground.

To make a simplified model of the most basic load the actuator will have to carry, the following free body diagram was created, in the plane normal to the rotational axis for the actuator (note the rectangular beam is representational, and may have a more complex geometry and mass distribution):

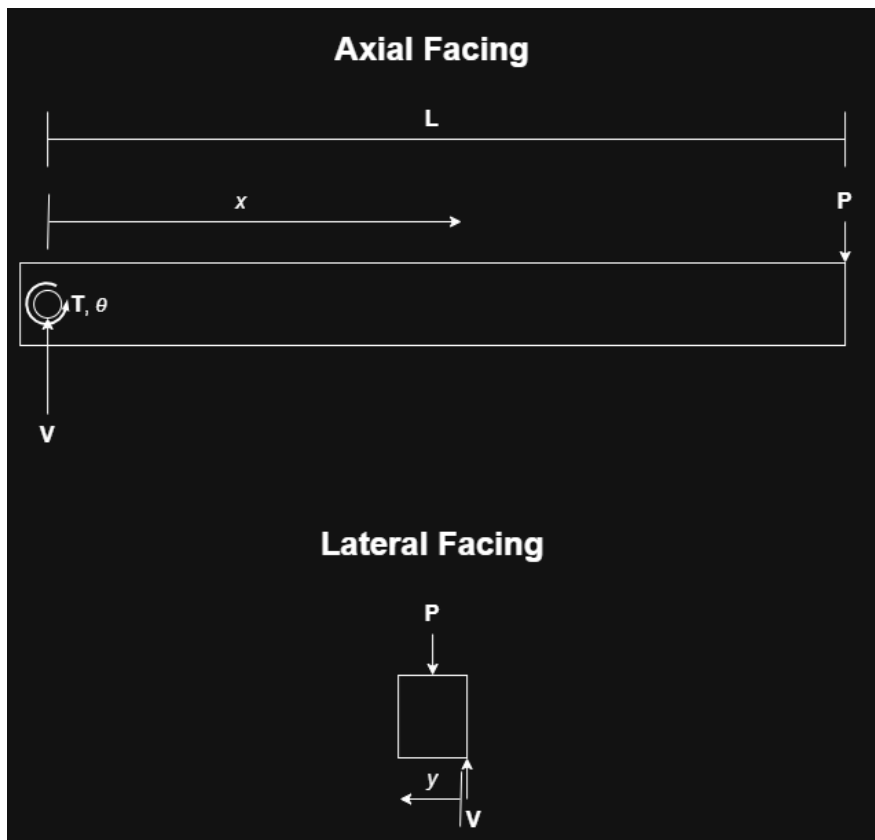


Figure 2: Simplified Load FBD

The parameter of interest is θ , which denotes the angular displacement of the actuator. The output torque of the actuator is denoted as T , and the radial reaction force of the bearing is denoted as V . Using the free body diagram, we can find the equation of motion of the actuator (disregarding its internal dynamics at this stage) and the statics of the radial load on the bearings.

$$J\ddot{\theta} = T - \int_0^L 9.8 \cdot m(x)x \cos \theta dx - PL$$

$$0 = V - P - 9.81M$$

$m(x)$ is the function of linkage mass at distance x , and M is the total mass of the linkage. The mass moment of inertia J , is a function of the properties of the beam, and may be defined as follows:

$$J = \int r_{\perp}^2 dm$$

We consider the idealised worst case, where the mass of the linkage is concentrated at its very end, and then compute the mass moment of inertia as well as the other mass dependent terms. According to NT-N-IRQ1, the mass would be 20kgs:

$$J = 20L^2$$

$$20L^2\ddot{\theta} = T - 20 \cdot 9.8 \cdot \cos \theta L - PL$$

$$T = 20L^2\ddot{\theta} + 20 \cdot 9.8 \cdot \cos \theta L + PL$$

$$T = 20L^2\ddot{\theta} + 20 \cdot 9.8 \cdot \cos \theta L + PL$$

With L being 0.25 metres, and assuming the very limit of capability being when acceleration is 0, the equation simplifies to:

$$T = 49 \cos \theta + 0.25P$$

Since NT-N-IRQ1 requires only a 20kg payload to be carried in frequencies comparable to human motion, we can assume that in the worst case, with $\cos \theta$ being 1, we will not be able to apply any external, torque-generating forces, represented by P . However, as $\cos \theta$ goes to 0, these external forces may increase up to 196N to match the force generated by a 20kg payload in the worst case.

Standard integrated actuators output generated power through a flange, with the output payload being fastened to the flange via machine screws. The force and load assessment of the flange is used to select the correct output transmission bearing.

From the analysis already done in the Axial Facing Plane, the radial load on the bearing will be at a maximum 196N. In the bottom of Figure 2 the Lateral Facing free body diagram is represented. In the simplified version of the free body diagram, if the beam is subject to the entire weight of the load (which is at a maximum 196N derived from NT-N-IRQ1) in the axial direction, then the output bearing must resist axial loads of up to 196N.

As P is applied with some offset y to the fastened plane on the right, there will be a moment created on the output bearing:

$$M = P \cdot y$$

As an assumption, the width of the lever arm described in NT-N-IRQ1 will not exceed its length, and is therefore capped at 0.25m (even this case is unrealistic). Therefore, the moment will be, at worst, as follows:

$$M = 196 \cdot 0.25M = 48Nm$$

Finally, the output speed is required for finalising output bearing choice. NT-N-IRQ1 dictates that the payload will move at motion frequencies of 20Hz, and it is assumed that the angular displacement of one cycle is at a maximum half a revolution. Thus, the calculation for angular velocity is as follows:

$$\omega = 20\pi \text{ rad} \cdot s^{-1}$$

The bearing must be able to resist these loads at 600RPM for 10,000 hours without failure (NT-N-IRQ1, NT-N-IRQ9, NT-N-IRQ13). Since the actuator should also be sealed from dust and high moisture (NT-N-IRQ10), the bearing should be sealed.

The output of the cycloidal disc is transmitted through bushings on the output rollers, which must additionally carry radial forces on top of the useful torque generation. These radial forces are dependent on the output torque, number of output rollers and diameter of the output rollers (including the bushings on top). From Li et al's work in [3], the following equilibrium describes the radial force of the bearings:

The bearing requirement summary is shown in Table 3.

Table 3: Bearing requirement summary

Requirement	Value	Basis
Axial load capacity	196N	Maximum payload from NT-N-IRQ1
Radial load capacity	196N	Equivalent radial reaction from payload
Overturning moment capacity	48Nm	$M = P \cdot y$ with $P = 196N$, $y = 0.25m$
Operating speed	600RPM	20Hz at half-revolution motion
Service life	10,000 hours	NT-N-IRQ9
Environmental protection	Sealed bearing	NT-N-IRQ10
Factor of safety	2	NT-N-IRQ13

3.1.5 Output Bearing Selection

Since the output interface is a flange, and considering all other requirements for the bearing, a crossed roller bearing with mounting holes is the most suitable architecture for the output bearing.

3.1.6 Cycloidal Reducer Design

Along with this, when the difference between the number of rollers and lobes is greater than 1, less than half of the lobes stay engaged during eccentric motion, even in the ideal case ([2]). With a difference of 1, every lobe is engaged (in ideal conditions) and the load on each individual lobe is drastically reduced. Therefore this will be the architecture used in this implementation.

REFERENCES

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